

BORDER, QC, Canada

WMO: 719010

Lat: **55.333N**Lon: **63.202W**Elev: **1572**StdP: **13.88**Time Zone: **-4.00 (NAA)**Period: **86-98**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.4	-29.1	-39.6	0.6	-31.7	-36.3	0.7	-28.7	47.9	-5.5	42.0	-5.5	10.5	270	0.952

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	71.4	56.7	67.6	55.1	63.0	52.7	59.0	66.7	56.3	62.6	54.1	60.5	11.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.9	70.5	60.6	53.4	64.3	58.4	51.0	58.7	56.7	26.5	66.3	24.7	63.1	23.3	60.2	66.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.9	31.2	26.8	DB	-38.7	79.0	2.1	5.8	-40.2	83.2	-41.4	86.6	-42.6	89.9	-44.1	94.1	(4)
(5)			WB	-38.0	63.1	3.2	2.0	-40.3	64.5	-42.2	65.7	-44.0	66.8	-46.3	68.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.5	-9.8	-9.8	3.1	19.0	32.1	43.5	51.2	50.1	41.2	30.1	16.8	0.0		
	DBStd	23.60	13.34	12.35	12.75	10.27	7.05	8.13	6.22	6.73	5.79	6.05	10.53	13.81		
	HDD50	10274	1853	1675	1453	930	554	229	62	82	272	616	997	1551		
	HDD65	15526	2318	2095	1918	1380	1019	647	428	462	715	1081	1447	2016		
	CDD50	229	0	0	0	0	1	35	99	87	7	0	0	0		
	CDD65	4	0	0	0	0	0	3	0	1	0	0	0	0		
	CDH74	70	0	0	0	0	0	43	8	19	0	0	0	0		
	CDH80	19	0	0	0	0	0	18	0	1	0	0	0	0		
	WSAvg	11.2	12.0	11.0	12.3	10.6	9.4	10.5	9.7	10.6	11.5	11.7	13.0	12.2		
	PrecAvg	29.6	2.0	1.7	1.7	1.8	1.9	2.8	4.3	3.2	3.3	2.8	2.2	1.8		
	PrecMax	38.6	4.1	4.0	4.4	4.7	5.4	4.3	7.3	5.5	6.8	5.2	3.8	5.6		
(15) Precipitation	PrecMin	20.3	0.2	0.1	0.4	0.3	0.4	0.8	1.9	1.7	1.1	1.6	0.4	0.3		
	PrecStd	4.1	0.9	1.1	1.1	0.9	1.0	0.8	1.2	0.9	1.2	0.9	0.8	1.2		
	DB	35.8	30.1	34.1	41.4	59.0	80.5	75.2	76.7	62.4	48.1	42.5	32.3			
	MCWB	35.4	29.7	31.4	35.8	47.8	61.3	59.9	61.0	50.7	44.5	42.2	32.0			
	2% DB	26.4	24.7	32.1	37.6	51.5	68.3	71.3	70.1	57.5	44.3	35.7	30.2			
(21) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	25.7	24.0	30.6	33.7	42.5	55.7	56.4	57.4	50.8	40.9	34.8	29.8			
	5% DB	20.9	17.9	29.8	35.6	46.2	62.9	67.7	66.2	53.7	41.0	32.4	25.2			
	MCWB	20.3	16.9	28.9	32.4	39.5	52.4	54.3	54.6	48.5	38.4	31.1	24.2			
	10% DB	12.3	10.7	24.6	33.5	41.4	57.6	64.0	62.3	50.3	37.7	30.5	21.1			
	MCWB	11.6	9.8	23.1	31.2	36.7	49.3	52.8	53.0	46.7	35.4	29.1	20.1			
	0.4% WB	32.4	29.1	32.3	36.9	47.9	61.4	61.2	63.0	54.9	45.6	41.6	31.5			
	MCDB	33.6	29.4	33.0	39.8	58.3	69.8	68.3	70.8	58.7	47.3	41.7	31.7			
	2% WB	25.3	23.7	30.8	34.7	43.1	56.7	58.2	59.2	51.7	41.5	34.6	29.3			
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	25.8	24.2	32.0	37.1	49.3	64.3	64.3	66.7	55.2	43.1	35.3	29.8			
	5% WB	19.5	17.0	27.9	32.7	40.0	53.4	56.1	56.3	49.1	38.2	31.5	24.5			
	MCDB	20.0	17.9	29.1	34.5	44.6	60.1	62.2	62.9	52.6	40.3	32.2	25.5			
	10% WB	11.4	9.8	22.8	31.3	37.4	50.1	53.9	54.1	47.0	35.6	29.3	20.1			
	MCDB	12.1	10.7	24.0	33.0	41.0	56.4	60.7	59.9	50.7	37.5	30.6	21.0			
(35) Mean Daily Temperature Range	MDBR	15.0	16.8	19.1	16.1	13.6	16.6	16.8	15.7	11.4	8.1	10.9	14.5			
	5% DB	23.3	21.0	18.5	16.4	20.9	27.4	24.6	23.1	17.3	11.8	11.3	14.3			
	MCWBR	23.4	20.9	17.7	13.4	12.9	14.1	10.8	10.7	11.5	9.5	11.6	14.4			
	5% WB	23.6	21.1	19.3	13.9	18.4	23.9	18.7	18.9	16.1	10.9	11.3	14.6			
	MCWBR	23.5	21.0	18.8	11.9	12.4	14.4	11.3	11.3	12.5	9.7	12.0	14.7			
(40) Clear-Sky Solar Irradiance	taub	0.227	0.242	0.268	0.332	0.392	0.368	0.378	0.354	0.329	0.284	0.240	0.222			
	taud	2.205	2.251	2.202	2.062	2.044	2.335	2.361	2.426	2.462	2.462	2.251	2.119			
	Ebn at Noon	231	272	288	280	268	278	272	273	266	254	222	198			
	Edh at Noon	17	25	33	45	49	37	36	31	26	20	16	15			
(44) All-Sky Solar Radiation	RadAvg	284	623	1134	1597	1735	1649	1458	1200	866	448	207	182			
	RadStd	24	59	101	121	171	150	104	110	75	37	36	23			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page